Parallel and distributed Computing

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Message passing interface (MPI) MIMD/SIMD

What is MPI?

Message Passing Interface (MPI)

- A standard for communication among processes in a distributed memory system
- Processes use send and receive operations
- Portable, scalable, and widely used in HPC (High-Performance Computing)

Key Features of MPI

- Allows multiple processes to run concurrently
- Each process has its own local memory
- Enables data sharing through messages
- Works well with heterogeneous systems
- Languages: C, C++, Fortran, Python (via mpi4py)

MIMD Architecture

Multiple Instruction, Multiple Data (MIMD)

- Each processor can run different code on different data
- Suitable for task-parallel systems
- Common in multi-core CPUs, clusters
- Used in MPI-based parallel applications
- 🖥 Example: Weather simulation running different models on different nodes

SIMD Architecture

Single Instruction, Multiple Data (SIMD)

- All processors execute the same instruction on different data
- Ideal for data-parallel operations
- Found in GPUs, vector processors
- Efficient for image processing, matrix operations, etc.
- 🖥 Example: Applying a filter to an image using a GPU

MPI with MIMD vs. SIMD

Feature	MIMD	SIMD
Instructions	Multiple	Single
Data	Multiple	Multiple
Communication	Often via MPI	Rare
Common in	CPU clusters	GPUs, vector cores
Use Case	Task parallelism	Data parallelism

Real-World Use Cases

MPI + MIMD:

- ◆ Climate modeling
- ◆ Large-scale simulations
- ◆ Financial modeling

SIMD:

- ◆ Image and video processing
- ◆ Deep learning (matrix multiplications)
- ◆ Scientific computing

Advice

Kindly read the book before and after the lecture, if some terms/ words are not covered in the lecture, if appear in exam it might not appear strange or out of course.